

Problem Set 2.1 (Bonus) – Climatic Risk Intelligence Module

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Source Code

```
1  """
2  Program: Climatic Risk Intelligence Module
3  Purpose: Classifies a work-site into a safety tier (Freeze Alert, Critical,
4           Cautionary, Operational) from a derived Heat Stress Index.
5  Author: Kshitiz
6
7  HSI = Temperature + (0.5 * Humidity). Freeze Alert is checked first
8  regardless of HSI, since a freezing site with high humidity can still
9  give a large HSI. Critical and Cautionary come next in that order, and
10 Operational is just the else case.
11 """
12
13 if (temp_str := input("Temperature (C): ").strip()):
14     try:
15         temperature = float(temp_str)
16     except ValueError:
17         temperature = None
18 else:
19     temperature = None
20
21 if temperature is None:
22     print("Safety Level: Unknown")
23 else:
24     humidity_str = input("Humidity (%): ").strip()
25     try:
26         humidity = float(humidity_str)
27         assert humidity >= 0, "Telemetry Error: Negative Humidity"
28     except ValueError:
29         humidity = None
30
31     if humidity is None:
32         print("Safety Level: Unknown")
33     else:
34         hsi = temperature + (0.5 * humidity)
35
36         if temperature <= 0:
37             tier = "FREEZE ALERT"
38         elif hsi > 45 or (temperature > 38 and humidity > 70):
39             tier = "CRITICAL"
40         elif 30 <= hsi <= 45:
41             wind_str = input("Wind speed (km/h): ").strip()
42             wind_speed = float(wind_str) if wind_str else 0.0
43             tier = "CAUTIONARY" if wind_speed < 5 else "OPERATIONAL"
44         else:
45             tier = "OPERATIONAL"
46
47     # Bonus: adjust Cautionary tier based on battery level
48     if tier == "CAUTIONARY":
49         battery_str = input("Battery level (%): ").strip()
50         battery = float(battery_str) if battery_str else 100.0
51         if battery < 20:
```

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52         tier = "CRITICAL"
53     elif battery > 80:
54         tier = "OPERATIONAL"
55
56     risk_label = "Safe" if hsi < 30 else "Unsafe"
57     print(f"HSI: {hsi:.1f}")
58     print(f"Safety Level: {tier}")
59     print(f"Risk Label: {risk_label}")

```

Execution Trace

```

1  --- Scenario (a): Empty input handled by walrus check ---
2  Temperature (C):
3  Safety Level: Unknown
4
5  --- Scenario (b): Critical heatwave reading ---
6  Temperature (C): 32
7  Humidity (%): 80
8  HSI: 72.0
9  Safety Level: CRITICAL
10 Risk Label: Unsafe
11
12 --- Scenario (c): Cautionary reading adjusted by Battery-Level bonus ---
13 Temperature (C): 25
14 Humidity (%): 20
15 Wind speed (km/h): 3
16 Battery level (%): 10
17 HSI: 35.0
18 Safety Level: CRITICAL
19 Risk Label: Unsafe
20
21 Note: primary tier for (25, 20, wind 3) is CAUTIONARY (30 <= HSI <= 45,
22 wind < 5). Battery level 10 is below the 20% threshold, so the Nested
23 Intelligence Suite elevates it to CRITICAL.

```

Submission Notes

Freeze Alert is checked before anything else because temperature alone decides it – a freezing site with high humidity can still produce a large HSI, and checking HSI-based tiers first would misclassify it as a heat emergency. Critical and Cautionary come next in that order, so by the time a Cautionary branch runs it already knows HSI is not above 45.

The empty-input check runs on the raw string from `input()`, guarded by the walrus operator, before any conversion happens. This matters because a temperature of 0 is falsy as a number but not falsy as the string "0", so checking the raw string avoids treating a genuine 0°C reading as missing input.